Blood performs the following functions:

1. Blood transports oxygen from lungs to the tissues and carbon dioxide from tissues to the lungs.
2. It transports absorbed nutrients from digestive tract to all the body tissues.
3. It transports metabolic waste materials to kidney, lungs, skin and intestine for their removal.
4. It transports various minerals, vitamins and hormones.
5. It regulates water balance.
6. It maintains acid-base balance in the body.
7. It provides defense against various infections through leukocytes and antibodies.

DO YOU KNOW?

Blood doping is a method followed to temporarily increase the oxygen carrying capacity of blood in an attempt to gain a competitive advantage by an athlete. In this method, blood is removed from the athlete, promptly reinfusing plasma but freezing the erythrocytes for reinfusion one to seven days before the competitive event. Blood doping is illegal in collegiate athletics and Olympics for ethical and medical reasons.

DO YOU KNOW?

The sales of synthetic erythropoietin (Epogen, Procrit) provides more than $1 billion dollars annually. This hormone is often used to boost erythrocyte production in patients with suppressed erythropoietic activity, such as those with kidney failure or those undergoing chemotherapy for cancer.

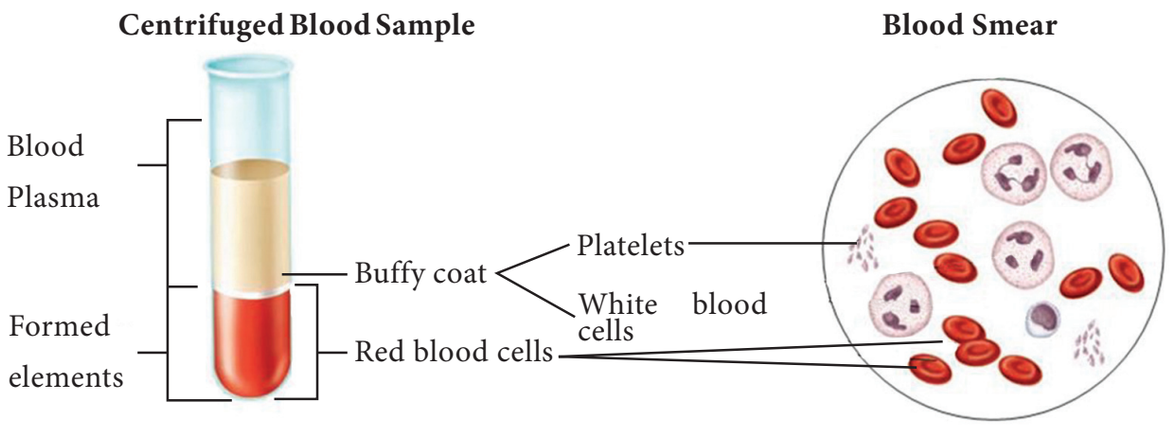


Figure 1. 18 Blood constituents; After centrifugation, blood cells settle at the bottom of the centrifuge tube leaving the plasma at the top of the tube. Leukocytes and thrombocytes form a thin light colored buffy coat at the interface between packed erythrocyte and the plasma.

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